

INTEGER-FLOOR LIFTING FOR FRANKL-COMPLETE UNIFORM CONFIGURATIONS

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ABSTRACT. Let $M_k(n)$ be the largest number of distinct k -sets that a Non-Frankl-complete collection supported on at most n points can contain. We prove the integer deletion inequality

$$M_k(n) \leq \left\lfloor \frac{n}{n-k} M_k(n-1) \right\rfloor.$$

Iterating the floor, rather than applying a seed bound only once, sharpens known local-configuration estimates. From the certified value $\text{FC}(4, 8) = 12$ we obtain the closed form

$$\text{FC}(4, n) \leq 1 + \left\lfloor \frac{\binom{n}{4} + n}{7} \right\rfloor \quad (n \geq 8),$$

including $\text{FC}(4, 9) \leq 20$ and $\text{FC}(4, 10) \leq 32$. We also describe an exact census of all 2,771,104 labelled union-closed families on five points. Its apparent frequency-power inequalities are shown to be artifacts of the bounded range: a twin-free power-set-chain construction violates every fixed nonnegative integer frequency power. All computational statements use integer arithmetic and have separate replay programs. These results concern local Frankl-complete configurations and do not prove Frankl's conjecture.

1. INTRODUCTION

A finite family $\mathcal{F}$ of sets is union-closed if $A \cup B \in \mathcal{F}$ whenever $A, B \in \mathcal{F}$. Frankl's conjecture, credited to Péter Frankl in 1979, asserts that every such family containing a nonempty set has an element in at least half of its members [2]. The conjecture remains open. Entropy and coupling arguments have established dimension-free constants, including the exact closed-form guarantee $(3 - \sqrt{5})/2$ [1] and the reported numerical bound 0.38234 [5]; neither reaches one half. Computation has verified the conjecture for ground sets of size at most twelve [4].

The local-configuration approach asks whether the presence of a fixed subcollection forces an abundant element. Poonen gave an exact weighting characterization of what are now called Frankl-complete collections [2]. Pulaj

Date: August 22, 2026.

2020 Mathematics Subject Classification. 05D05, 05A20, 68V20.

Key words and phrases. union-closed sets, Frankl's conjecture, Frankl-complete family, integer averaging, exact enumeration.

and Wood recently determined several uniform thresholds, including

$$\text{FC}(4, 8) = 12, \quad \text{FC}(5, 7) = 14, \quad \text{FC}(6, 8) = 26,$$

and converted them into general bounds by averaging a certified seed over subsets of a larger ground set [3]. For instance, their direct seed-to- n estimate gives $\text{FC}(4, 9) \leq 21$ and has leading density $11/70$ relative to $\binom{n}{4}$.

Our first result is a small but systematic strengthening: apply the averaging argument one point at a time and retain every integer floor. The resulting recurrence has a closed solution for four-sets and leading density $1/7$. The proof is a symmetric Farkas certificate: add the n point-deletion constraints with equal multipliers.

The second part records a deliberately adversarial use of exact enumeration. Several low-complexity weighted inequalities survive all families on five points, but a symbolic construction then disproves them in every fixed degree. This prevents bounded computational evidence from being mistaken for a route to the full conjecture.

2. FRANKL-COMPLETE COLLECTIONS AND DELETION

Definition 2.1. A collection $\mathcal{A}$ of sets is *Frankl-complete* (FC) if every union-closed family containing $\mathcal{A}$ has an element of $U(\mathcal{A}) = \bigcup_{A \in \mathcal{A}} A$ in at least half of its members. It is *Non-FC* otherwise. For $k \geq 1$, let $M_k(n)$ be the maximum size of a Non-FC collection of distinct k -sets whose support has at most n points. Equivalently, whenever the threshold is defined,

$$\text{FC}(k, n) = M_k(n) + 1.$$

FC is monotone: a collection containing an FC subcollection is FC. The support-at-most convention is important below, since a deletion may leave fewer than $n - 1$ active points.

Theorem 2.2 (integer deletion lift). *Let $n > k$. If $M_k(n - 1) \leq a$, then*

$$(2.1) \quad M_k(n) \leq \left\lfloor \frac{na}{n - k} \right\rfloor.$$

Proof. Let $\mathcal{H}$ be a Non-FC collection of distinct k -sets supported on at most n points. The assertion is immediate if its support has at most $n - 1$ points, so assume that its support is $[n]$. For $v \in [n]$, set

$$\mathcal{H} - v = \{E \in \mathcal{H} : v \notin E\}.$$

If $\mathcal{H} - v$ were FC, monotonicity would make $\mathcal{H}$ FC. Hence every $\mathcal{H} - v$ is Non-FC and $|\mathcal{H} - v| \leq a$. Summing these n inequalities, each k -set is counted once for each point outside it, and therefore

$$(n - k)|\mathcal{H}| = \sum_{v=1}^n |\mathcal{H} - v| \leq na.$$

Taking the integer floor proves (2.1). □

The equal coefficients in this sum are precisely the automorphism-averaged dual multipliers of the point-deletion constraints. Iteration gives the following consequence.

Corollary 2.3. *Suppose $M_k(n_0) \leq a_{n_0}$, and define*

$$(2.2) \quad a_n = \left\lfloor \frac{n}{n-k} a_{n-1} \right\rfloor \quad (n > n_0).$$

Then $M_k(n) \leq a_n$, and consequently $\text{FC}(k, n) \leq a_n + 1$.

Repeated adjacent floors can only improve a direct lift from an earlier support size: the product of the unfloored adjacent ratios is $\binom{n}{k}/\binom{j}{k}$. Thus no earlier bound generated by the same seed is lost by using (2.2).

3. A CLOSED FOUR-SET BOUND

The known values $\text{FC}(4, 5) = 5$, $\text{FC}(4, 6) = 7$, $\text{FC}(4, 7) = 10$, and $\text{FC}(4, 8) = 12$ [3] imply $M_4(8) \leq 11$ under the support-at-most convention. Put $a_8 = 11$ and use (2.2).

Theorem 3.1. *For every $n \geq 8$,*

$$a_n = \left\lfloor \frac{\binom{n}{4} + n}{7} \right\rfloor.$$

In particular,

$$(3.1) \quad \boxed{\text{FC}(4, n) \leq 1 + \left\lfloor \frac{\binom{n}{4} + n}{7} \right\rfloor} \quad (n \geq 8).$$

Proof. The formula gives $a_8 = 11$. Write

$$Q_n = \binom{n}{4} + n = 7A_n + s_n, \quad 0 \leq s_n < 7.$$

For residues $n \equiv 0, 1, \dots, 6 \pmod{7}$, direct reduction gives

$$(3.2) \quad (s_n) = (0, 1, 2, 3, 5, 3, 0).$$

Assuming $a_{n-1} = A_{n-1}$ and using $\binom{n}{4} = \frac{n}{n-4} \binom{n-1}{4}$, one obtains

$$(3.3) \quad \frac{n}{n-4} A_{n-1} = A_n + \frac{s_n}{7} + \frac{n(3-s_{n-1})}{7(n-4)}.$$

For the seven residue classes, the term following A_n in (3.3) is, respectively,

$$(3.4) \quad \begin{aligned} & \frac{3n}{7(n-4)}, \quad \frac{1}{7} + \frac{3n}{7(n-4)}, \quad \frac{2}{7} + \frac{2n}{7(n-4)}, \quad \frac{3}{7} + \frac{n}{7(n-4)}, \\ & \frac{5}{7}, \quad \frac{3}{7} - \frac{2n}{7(n-4)}, \quad 0. \end{aligned}$$

The first possible $n \geq 9$ in these classes is 14, 15, 9, 10, 11, 12, 13, respectively. Each expression in (3.4) lies in $[0, 1)$ at that first value and thereafter. Taking floors in (3.3) proves the induction, and Corollary 2.3 gives (3.1). $\square$

The first values and the corresponding direct consequences of the Pulaj–Wood seed are shown in Table 1.

TABLE 1. Four-set FC upper bounds. The last column is the published direct seed-to- n ceiling bound.

n	recursive bound	direct bound
8	12	12
9	20	21
10	32	34
11	49	53
12	73	79
13	105	114
14	146	159
15	198	216

The leading coefficient improves from $11/70$ to $1/7$ relative to $\binom{n}{4}$. Other certified seeds immediately give

$$\text{FC}(5, 8) \leq 35, \quad \text{FC}(5, 9) \leq 77, \quad \text{FC}(6, 10) \leq 188.$$

4. A NO-GO THEOREM FOR FIXED FREQUENCY POWERS

For a union-closed family $\mathcal{F}$, put $m = |\mathcal{F}|$ and let f_x be the frequency of x . A natural candidate suggested by bounded searches is

$$(4.1) \quad W_p(\mathcal{F}) = \sum_{x \in U(\mathcal{F})} f_x^p (2f_x - m) \geq 0.$$

If (4.1) held with nonnegative weights, it would imply an abundant element. The following construction rules out this entire fixed-power class, even after twin elements (equal incidence columns) are removed.

Theorem 4.1. *For every fixed integer $p \geq 0$, there is a finite twin-free union-closed family $\mathcal{F}$ for which $W_p(\mathcal{F}) < 0$.*

Proof. Let K and $C = \{c_1, \dots, c_q\}$ be disjoint, with $|K| = k$, and set

$$(4.2) \quad \mathcal{G}_{k,q} = \mathcal{P}(K) \cup \{K \cup \{c_1, \dots, c_i\} : 1 \leq i \leq q\}.$$

This is the union of a power set and a chain, and it is union-closed. It has $2^k + q$ members. Every point of K has frequency $2^{k-1} + q$, while the frequencies on C are $1, 2, \dots, q$. Hence

$$(4.3) \quad W_p(\mathcal{G}_{k,q}) = k(2^{k-1} + q)^p q + \sum_{i=1}^q i^p (2i - 2^k - q).$$

Take $q = 2^k = t$. The power-sum estimate $\sum_{i=1}^t i^d = t^{d+1}/(d+1) + O_d(t^d)$ transforms (4.3) into

$$W_p(\mathcal{G}_{k,t}) = -\frac{2}{(p+1)(p+2)} t^{p+2} + O_p((k+1)t^{p+1}).$$

Because $t = 2^k$, this is negative for all sufficiently large k . Distinct points of K have distinct columns on $\mathcal{P}(K)$; points of C have distinct frequencies at most t , whereas the old-point frequency is $3t/2$. Thus the family is twin-free. $\square$

The obstruction occurs early. For $(p, k, q) = (1, 5, 7)$, the family has 12 points and 39 sets, with frequencies 23 five times and $1, \dots, 7$ once each. Direct substitution gives

$$5 \cdot 23(46 - 39) + \sum_{i=1}^7 i(2i - 39) = -7.$$

The family nevertheless satisfies Frankl's conclusion through each point of K ; Theorem 4.1 invalidates only the proposed averaging method.

5. EXACT FIVE-POINT CENSUS

For discovery and falsification, we enumerated all labelled union-closed families on five points, including the empty family. A family is split as $(\mathcal{A}, \mathcal{B})$ according to whether the fifth point is absent or present. Both traces are union-closed families on four points, and the sole cross condition is

$$(5.1) \quad a \cup b \in \mathcal{B} \quad (a \in \mathcal{A}, b \in \mathcal{B}).$$

There are 4960 possible four-point traces. Testing (5.1) gives exactly 2,771,104 five-point families. Table 2 records the outcome for (4.1); “twin-reduced” means that one term is retained per nonempty incidence-column class.

TABLE 2. Exact weighting audit over all five-point families with nonempty support.

p	all active elements		twin-reduced	
	negative count	minimum	negative count	minimum
0	14,020	-11	12,165	-11
1	65	-4	0	0
2	0	0	0	0
3	0	0	0	0

The apparent success for $p = 2, 3$, and for twin-reduced $p = 1$, is fully explained by the bounded range: Theorem 4.1 supplies exact failures beyond it. This census therefore contributes no new verified ground-set range beyond the known twelve-point result [4].

The primary enumerator represents the valid choices of $\mathcal{B}$ as 4960-bit integers. A separately written replay checks all 4960^2 trace pairs directly. The minimum Frankl margin $2 \max_x f_x - m$ is zero, and 202 labelled families attain equality. No floating-point decision is used.

6. SCOPE AND REPRODUCIBILITY

Theorem 3.1 is a local-configuration upper bound; it neither proves Frankl’s conjecture nor improves the universal frequency constant. The recorded literature search (completed August 22, 2026) located the direct predecessor [3], but no statement of the recursive closed form (3.1). This is a bounded priority search, not proof that the formula has never appeared in an unindexed note or computation.

The accompanying project contains the theorem proof, exact JSON tables, standard-library Python generators, independent replay programs, and shell test targets. The full five-point replay tests all trace pairs and is separated from the fast arithmetic checks.

COMPUTATIONAL AND AI DISCLOSURE

OpenAI Codex was used for literature triage, proof exploration, code drafting, adversarial testing, and language editing. All mathematical claims were reduced to the proofs or exact integer computations given here; the author checked the final arguments and takes responsibility for the manuscript.

REFERENCES

1. Ryan Alweiss, Brice Huang, and Mark Sellke, *Improved lower bound for Frankl’s union-closed sets conjecture*, The Electronic Journal of Combinatorics **31** (2024), no. 3, P3.35.
2. Bjorn Poonen, *Union-closed families*, Journal of Combinatorial Theory, Series A **59** (1992), no. 2, 253–268.
3. Jonad Pulaj and Kenan Wood, *Local configurations in union-closed families*, Experimental Mathematics **34** (2024), no. 4, 719–727.
4. Bojan Vučković and Miodrag Živković, *The 12-element case of Frankl’s conjecture*, IPSI BgD Transactions on Internet Research **13** (2017), no. 1, 65–71.
5. Lei Yu, *Dimension-free bounds for the union-closed sets conjecture*, Entropy **25** (2023), no. 5, 767.

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